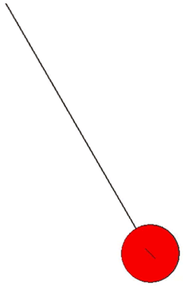
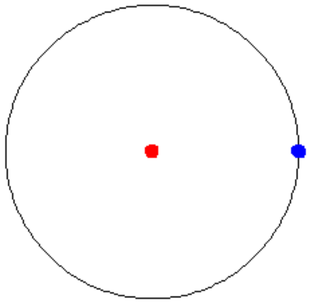
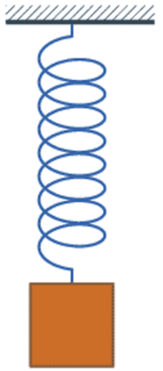
Basic Analysis of Vibrations

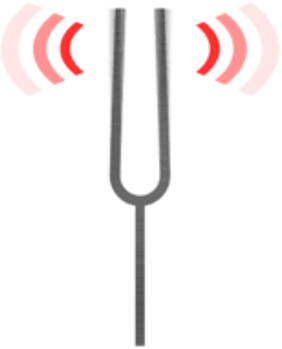
D. H. S. Maithripala, PhD
Dept. of Mechanical Engineering
University of Peradeniya

What do we mean by vibration







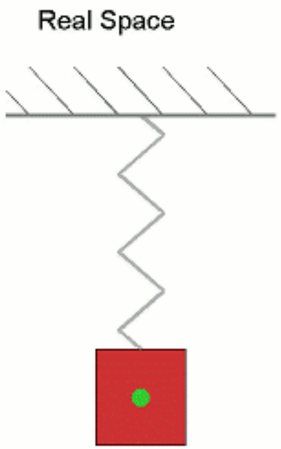


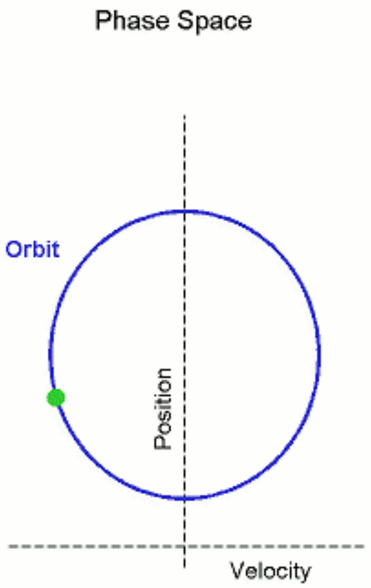
All of these are periodic motions…

Which one do you associate with the word Vibration ?

Typically Vibrations deal with High Frequency Oscillations

Simple Harmonic Motion





Newton's Laws

$$m\frac{d^2x}{dt^2} = -kx$$

$$x(t) = A\sin(\omega t + \varphi)$$

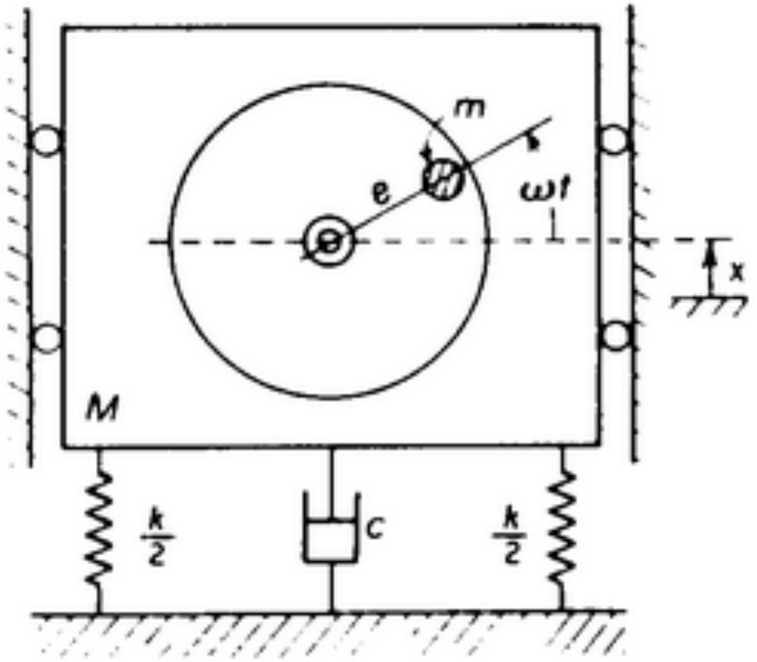
Physical Behavior ⟷ **Law of nature** ⟷ **Mathematical Concept**

The art of Science

Typical Source of Vibrations



Rotary Machine



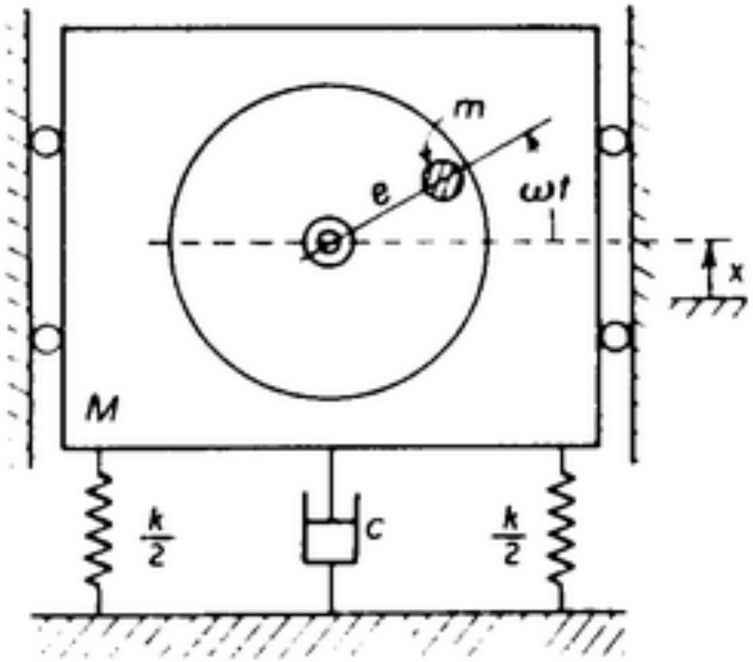
$$F_x = me\omega^2 \sin \omega t$$

Vertical Force felt by the foundation

Simple Modelling of Vibrations



Rotary Machine



$$(m+M)\ddot{x} = -kx - c\dot{x} + me\omega^2 \sin \omega t$$

Newton's Law

$$\omega_n^2 \triangleq \frac{k}{m+M} \qquad \zeta \triangleq \frac{1}{2\omega_n}\frac{c}{m+M}$$

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = f(t)$$

$$f(t) \triangleq \frac{me\omega^2}{(m+M)} \sin \omega t$$

$$F_x = me\omega^2 \sin \omega t$$

Vertical Force felt by the foundation

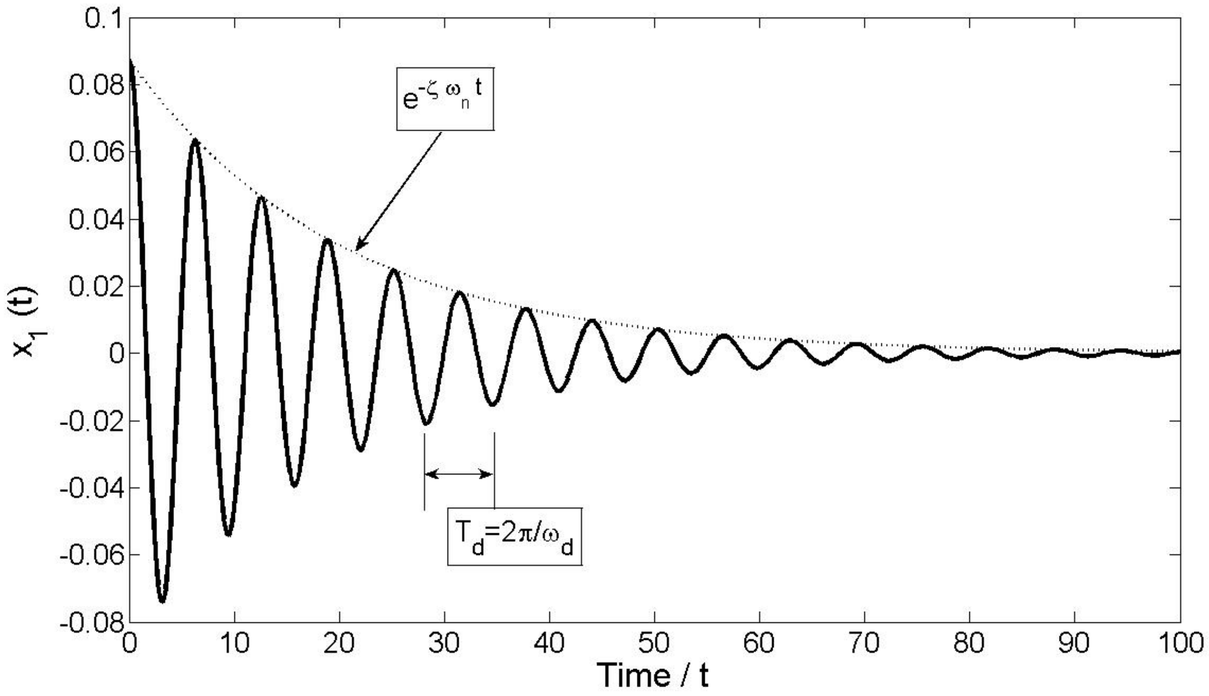
Simple Free Vibrations

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = f(t)$$

$$f(t) \equiv 0 \qquad 0 \le \zeta < 1$$

$$x_{ic}(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

$$\omega_d \triangleq \omega_n\sqrt{1-\zeta^2}$$

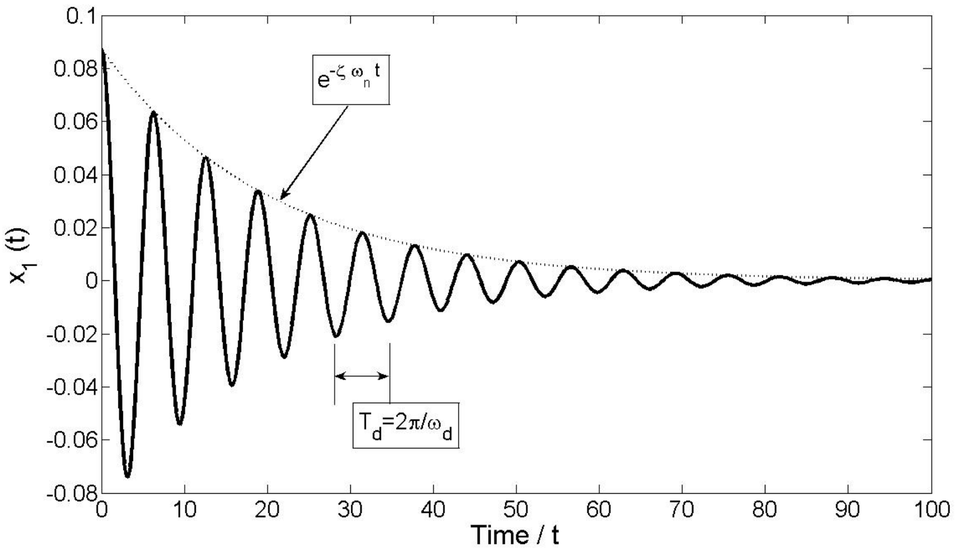


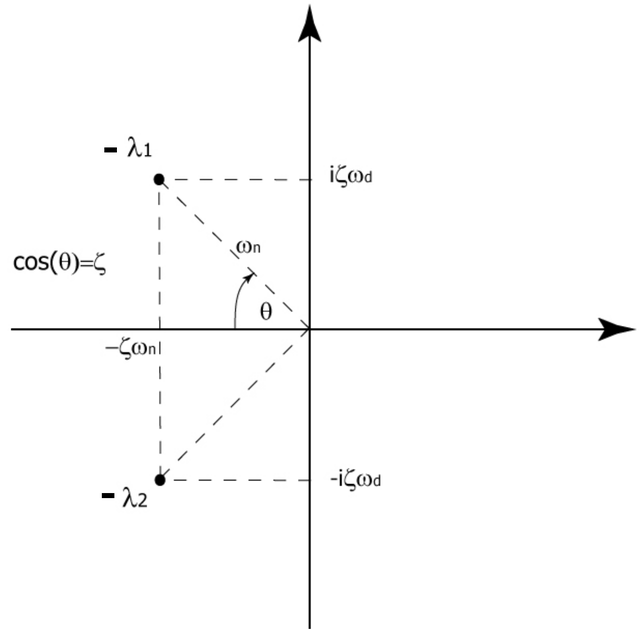
Simple Free Vibrations

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = 0$$

$$0 \le \zeta < 1 \qquad \omega_d \triangleq \omega_n\sqrt{1-\zeta^2}$$

Characteristic Polynomial $\Delta(s) = s^2 + 2\zeta\omega_n s + \omega_n^2 = (s+\lambda_1)(s+\lambda_2)$





$$x_{ic}(t) = Ae^{-\zeta\omega_n t}\sin(\omega_d t + \phi)$$

Simple Forced Vibrations

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \frac{F_0}{M}\cos\omega t$$

$$0 \leq \zeta < 1$$

Laplace Transform

$$X(s) = \underbrace{\frac{1/M}{s^2 + 2\zeta\omega_n s + \omega_n^2}}_{G(s)} \left(\frac{F_0\, s}{s^2 + \omega^2} \right)$$

Inverse Laplace Transform

$$x_{ss}(t) = \chi(\omega)\, F_0\, \cos(\omega t + \phi(\omega))$$

$$\chi(\omega) = \left| G(i\omega) \right| = \frac{1/M}{\sqrt{(\omega_n^2 - \omega^2)^2 + 4\zeta^2\omega^2\omega_n^2}}$$

$$\phi(\omega) = \angle G(i\omega) = \arctan\left(\frac{2\zeta\omega\omega_n}{\omega_n^2 - \omega^2} \right)$$

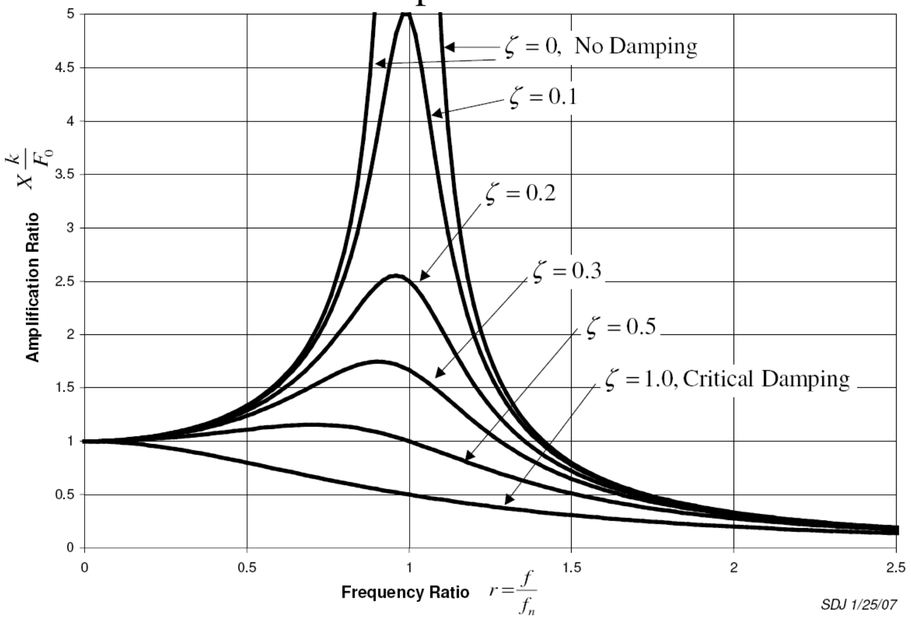
Simple Forced Vibrations

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \frac{F_0}{M}\cos\omega t$$

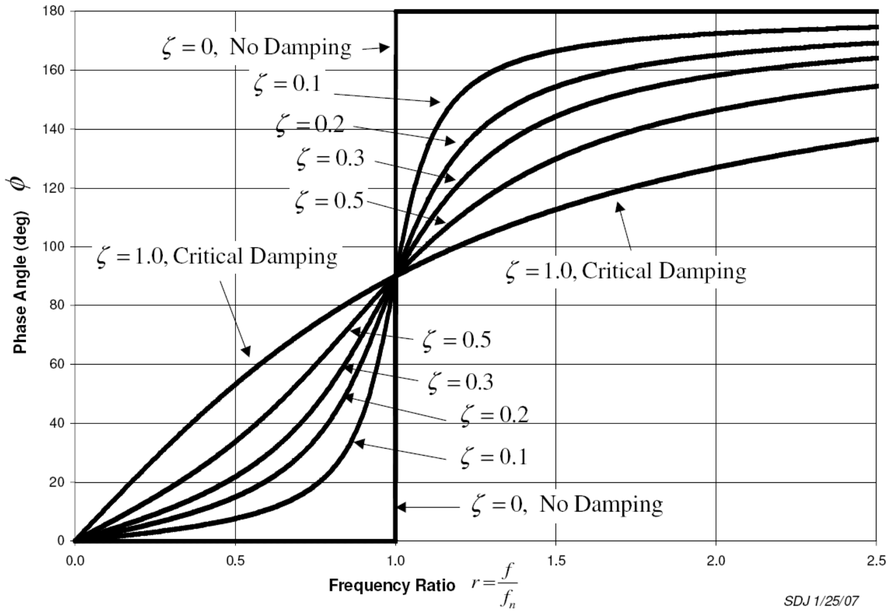
$$0 \le \zeta < 1$$

$$x_{ss}(t) = \chi(\omega)\, F_0\, \cos(\omega t + \phi(\omega))$$

Amplitude



Phase



$$\omega_r = \omega_n\sqrt{1-2\zeta^2}$$

Resonance Frequency

$$x_{ss_{max}} = \frac{F_0/M}{2\zeta\omega_n^2\sqrt{1-\zeta^2}}$$

Resonance Amplitude

$$\chi(\omega) = \frac{1/M}{\sqrt{(\omega_n^2-\omega^2)^2 + 4\zeta^2\omega^2\omega_n^2}}$$

$$\phi(\omega) = \arctan\left(\frac{2\zeta\omega\omega_n}{\omega_n^2-\omega^2}\right)$$

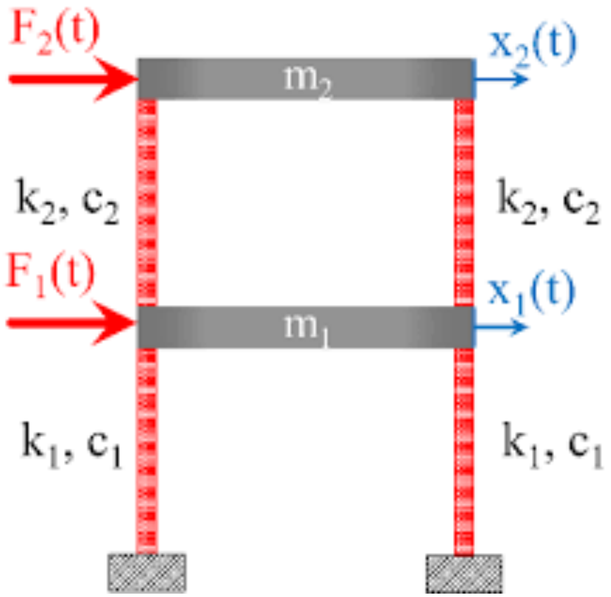
Analysis of Vibrations
2-DOF Systems

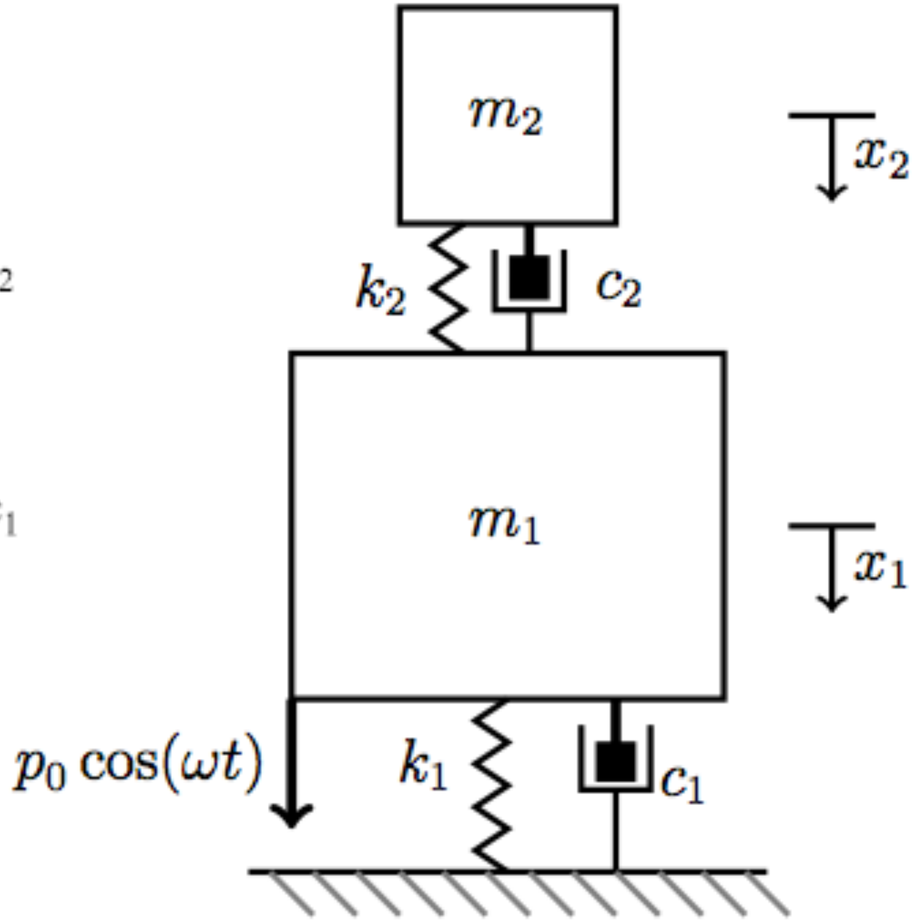
D. H. S. Maithripala, PhD
Dept. of Mechanical Engineering
University of Peradeniya

2-DOF Vibrations

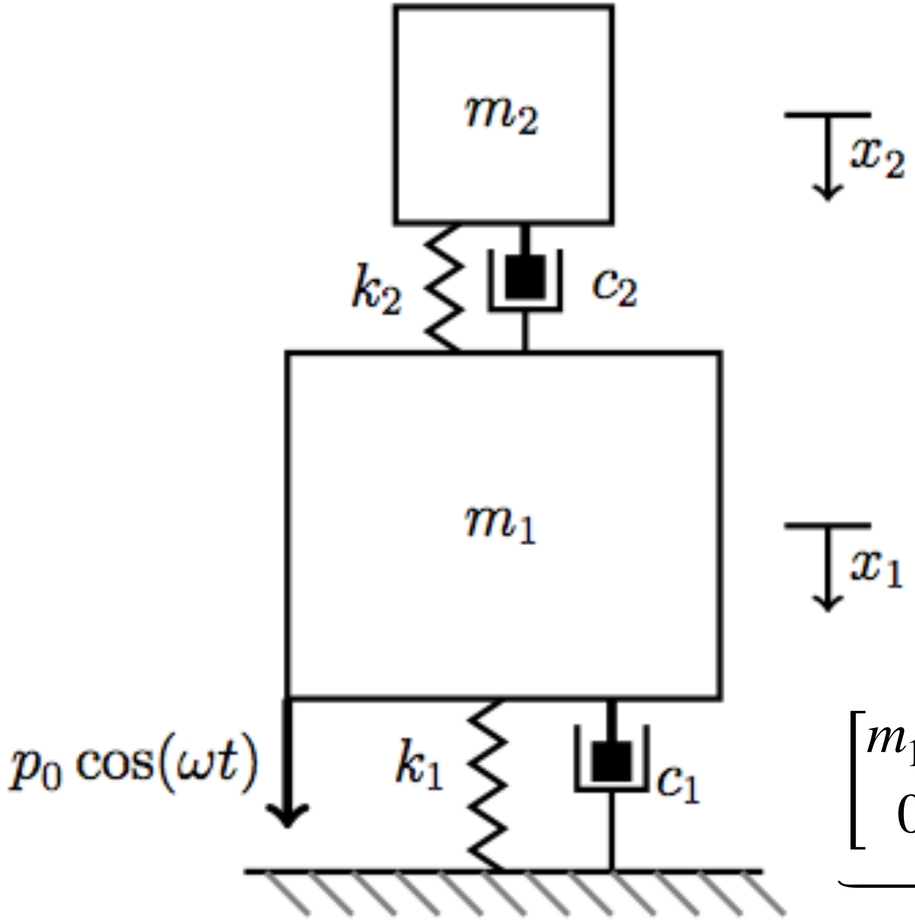


Rotary Machine





2-DOF Vibrations



With Reference Equilibrium Points

$$m_1\ddot{x}_1 + (c_1 + c_2)\dot{x}_1 + (k_1 + k_2)x_1 - c_2\dot{x}_2 - k_2x_2 = f_1(t)$$
$$m_2\ddot{x}_2 + c_2\dot{x}_2 + k_2x_2 - c_2\dot{x}_1 - k_2x_1 = 0$$

$$\underbrace{\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix}}_{M} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \underbrace{\begin{bmatrix} (c_1 + c_2) & -c_2 \\ -c_2 & c_2 \end{bmatrix}}_{C} \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} + \underbrace{\begin{bmatrix} (k_1 + k_2) & -k_2 \\ -k_2 & k_2 \end{bmatrix}}_{K} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} f_1(t) \\ 0 \end{bmatrix}$$

$$M\ddot{x} + C\dot{x} + Kx = f(t) \qquad x \triangleq \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

2-DOF Vibrations

For Zero Initial Conditions

$$\mathscr{L}\{M\ddot{x} + C\dot{x} + Kx\} = \mathscr{L}\{f(t)\}$$

$$(Ms^2 + Cs + K)X(s) = F(s)$$

$$X(s) \triangleq \mathscr{L}\{x(t)\} \qquad F(s) \triangleq \mathscr{L}\{f(t)\}$$

$$\Delta(s) \triangleq (Ms^2 + Cs + K) = \begin{bmatrix} (m_1 s^2 + (c_1 + c_2)s + (k_1 + k_2)) & -(c_2 s + k_2) \\ -(c_2 s + k_2) & (m_2 s^2 + c_2 s + k_2) \end{bmatrix}$$

$$X(s) = \Delta(s)^{-1} F(s) = \begin{bmatrix} G_1(s)F_1(s) \\ G_2(s)F_1(s) \end{bmatrix}$$

$$G_1(s) = \frac{m_2 s^2 + c_2 s + k_2}{|\Delta(s)|} \qquad G_2(s) = \frac{c_2 s + k_2}{|\Delta(s)|}$$

$$F_1(s) = \mathscr{L}\{f_1(t)\}$$

$$|\Delta(s)| = \det(\Delta(s)) = \left(m_1 s^2 + (c_1 + c_2)s + (k_1 + k_2)\right)(m_2 s^2 + c_2 s + k_2) - (c_2 s + k_2)^2$$

2-DOF Vibrations

For Zero Initial Conditions

$$\mathscr{L}\{M\ddot{x} + C\dot{x} + Kx\} = \mathscr{L}\{f(t)\}$$

$$(Ms^2 + Cs + K)X(s) = F(s)$$

$$G_1(s) = \frac{m_2 s^2 + c_2 s + k_2}{|\Delta(s)|} \qquad G_2(s) = \frac{c_2 s + k_2}{|\Delta(s)|}$$

$$F_1(s) = \mathscr{L}\{f_1(t)\}$$

$$X(s) = \Delta(s)^{-1}F(s) = \begin{bmatrix} G_1(s)F_1(s) \\ G_2(s)F_1(s) \end{bmatrix} \Rightarrow \begin{bmatrix} x_1(t) \\ x_2(t)) \end{bmatrix} = \mathscr{L}^{-1}\left\{ \begin{bmatrix} G_1(s)F_1(s) \\ G_2(s)F_1(s) \end{bmatrix} \right\} = \begin{bmatrix} \mathscr{L}^{-1}\{G_1(s)F_1(s)\} \\ \mathscr{L}^{-1}\{G_2(s)F_1(s)\} \end{bmatrix}$$

$$|\Delta(s)| = \det(\Delta(s)) = \left(m_1 s^2 + (c_1 + c_2)s + (k_1 + k_2)\right)(m_2 s^2 + c_2 s + k_2) - (c_2 s + k_2)^2$$

2-DOF Vibrations

$$X(s) = \Delta(s)^{-1}F(s) = \begin{bmatrix} G_1(s)F_1(s) \\ G_2(s)F_1(s) \end{bmatrix} \qquad \begin{bmatrix} x_1(t) \\ x_2(t)) \end{bmatrix} = \mathscr{L}^{-1}\left\{ \begin{bmatrix} G_1(s)F_1(s) \\ G_2(s)F_1(s) \end{bmatrix} \right\} = \begin{bmatrix} \mathscr{L}^{-1}\left\{G_1(s)F_1(s)\right\} \\ \mathscr{L}^{-1}\left\{G_2(s)F_1(s)\right\} \end{bmatrix}$$

$$F_1(s) = \mathscr{L}\{f_1(t)\} \qquad f_1(t) = p_0\cos(\omega t)$$

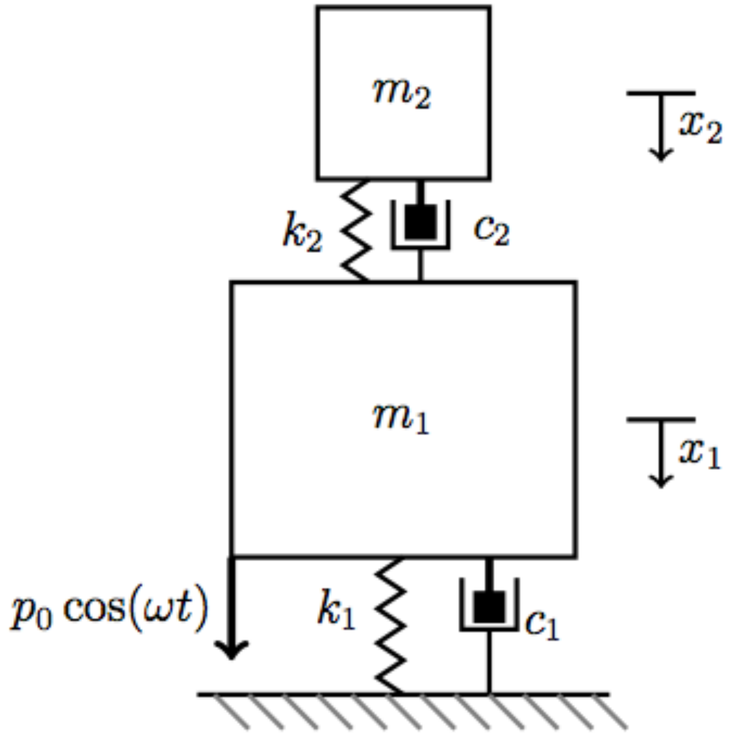
$$\begin{bmatrix} x_1(t) \\ x_2(t)) \end{bmatrix} = \begin{bmatrix} \chi_1(\omega)p_0\cos(\omega t + \phi_1(\omega)) \\ \chi_2(\omega)p_0\cos(\omega t + \phi_2(\omega)) \end{bmatrix} \qquad \begin{aligned} \chi_1(\omega) &= |\,G_1(i\omega)\,| \qquad & \phi_1(\omega) &= \angle G_1(i\omega) \\ \chi_2(\omega) &= |\,G_2(i\omega)\,| \qquad & \phi_2(\omega) &= \angle G_2(i\omega) \end{aligned}$$

$$G_1(s) = \frac{m_2s^2 + c_2s + k_2}{|\,\Delta(s)\,|} \qquad G_2(s) = \frac{c_2s + k_2}{|\,\Delta(s)\,|}$$



$$|\,\Delta(s)\,| = \det(\Delta(s)) = \Big(m_1s^2 + (c_1 + c_2)s + (k_1 + k_2)\Big)(m_2s^2 + c_2s + k_2) - (c_2s + k_2)^2$$

Coupled 2-DOF Vibrations



$$x(t) = \begin{bmatrix} \chi_1(\omega) p_0 \cos(\omega t + \phi_1(\omega)) \\ \chi_2(\omega) p_0 \cos(\omega t + \phi_2(\omega)) \end{bmatrix}$$

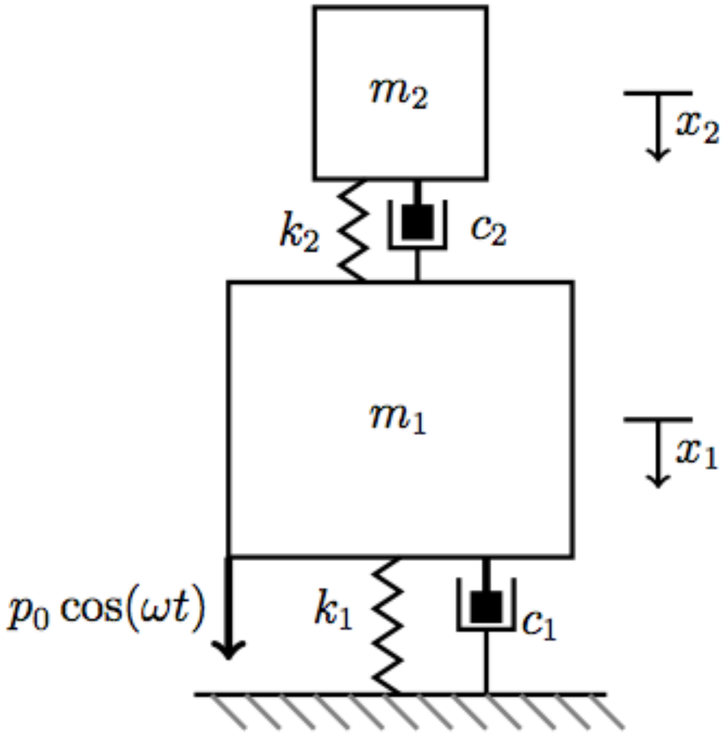
$$\chi_1(\omega) = |G_1(i\omega)|, \qquad \chi_2(\omega) = |G_2(i\omega)|$$
$$\phi_1(\omega) = \angle G_1(i\omega), \qquad \phi_2(\omega) = \angle G_2(i\omega),$$

$$\omega_{n1}^2 = \frac{k_1}{m_1}, \quad \omega_{n2}^2 = \frac{k_2}{m_2}, \quad 2\zeta_1\omega_{n1} = \frac{c_1}{m_1}, \quad 2\zeta_2\omega_{n2} = \frac{c_2}{m_2}, \quad \mu = \frac{m_2}{m_1}$$

$$G_1(s) = \frac{\frac{1}{m_1}(s^2 + 2\zeta_2\omega_{2n}s + \omega_{2n}^2)}{|\Delta(s)|} \qquad G_2(s) = \frac{\frac{1}{m_1}(2\zeta_2\omega_{2n}s + \omega_{2n}^2)}{|\Delta(s)|}$$

$$|\Delta(s)| = \det\left(\begin{bmatrix} \left(s^2 + (2\zeta_1\omega_{n1} + 2\mu\zeta_2\omega_{n2})s + (\omega_{n1}^2 + \mu\omega_{n2}^2)\right) & -\mu(2\zeta_2\omega_{n2}s + \omega_{n2}^2) \\ -(2\zeta_2\omega_{n2}s + \omega_{n2}^2) & (s^2 + 2\zeta_2\omega_{n2}s + \omega_{n2}^2) \end{bmatrix} \right)$$

Coupled 2-DOF Vibrations



$$x(t) = \begin{bmatrix} \chi_1(\omega) p_0 \cos(\omega t + \phi_1(\omega)) \\ \chi_2(\omega) p_0 \cos(\omega t + \phi_2(\omega)) \end{bmatrix}$$

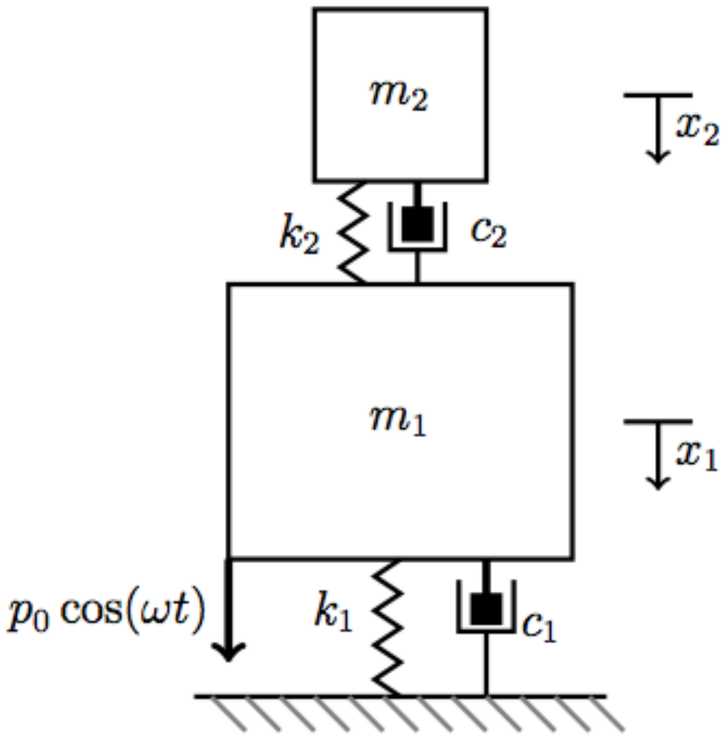
$$\chi_1(\omega) = |G_1(i\omega)|, \qquad \chi_2(\omega) = |G_2(i\omega)|$$
$$\phi_1(\omega) = \angle G_1(i\omega), \qquad \phi_2(\omega) = \angle G_2(i\omega),$$

$$\zeta_1 = 0 \quad \zeta_2 = 0$$

$$G_1(s) = \frac{1}{m_1} \frac{(s^2 + \omega_{2n}^2)}{|\Delta(s)|}, \qquad G_2(s) = \frac{1}{m_1} \frac{\omega_{2n}^2}{|\Delta(s)|}$$

$$|\Delta(s)| = \det\left(\begin{bmatrix} \left(s^2 + (\omega_{n1}^2 + \mu\omega_{n2}^2)\right) & -\mu\omega_{n2}^2 \\ -\omega_{n2}^2 & (s^2 + \omega_{n2}^2) \end{bmatrix} \right) = \left(s^2 + (\omega_{n1}^2 + \mu\omega_{n2}^2)\right)(s^2 + \omega_{n2}^2) - \mu\omega_{n2}^4$$

Coupled 2-DOF Vibrations



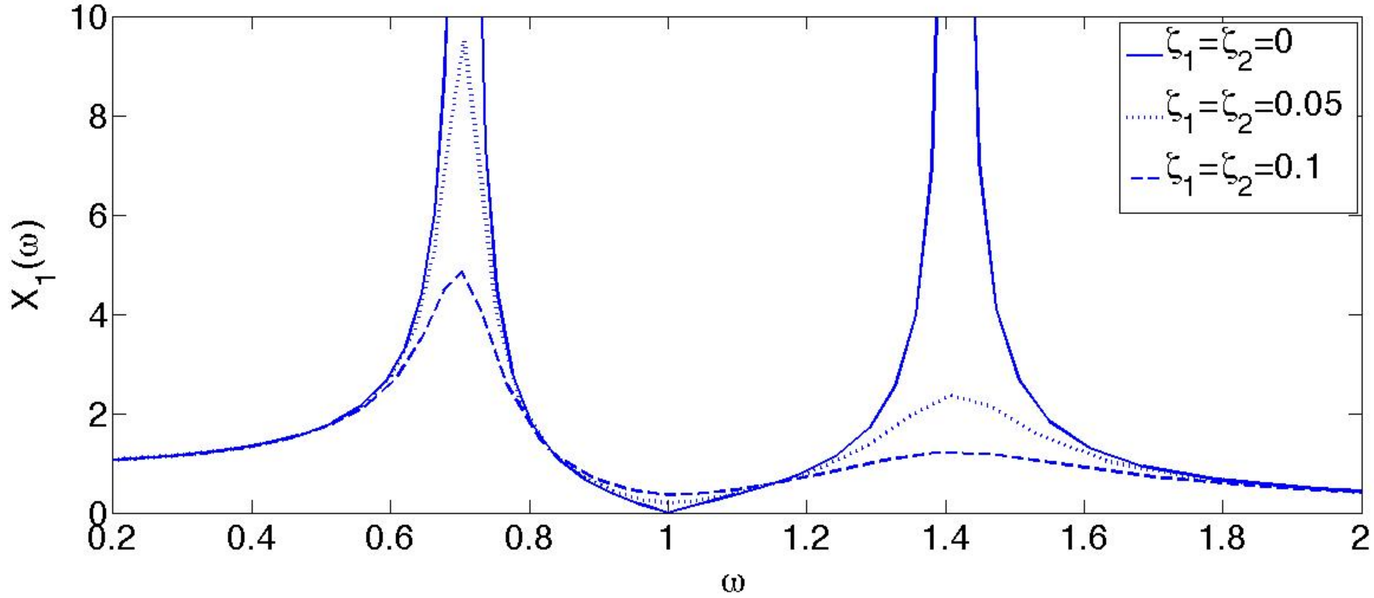
$$\chi_1(\omega) = |G_1(i\omega)| \qquad \chi_2(\omega) = |G_2(i\omega)|$$
$$\phi_1(\omega) = \angle G_1(i\omega) \qquad \phi_2(\omega) = \angle G_2(i\omega)$$

$$\zeta_1 = 0 \quad \zeta_2 = 0$$

$$G_1(i\omega) = \frac{1}{m_1}\frac{(-\omega^2 + \omega_{2n}^2)}{|\Delta(i\omega)|} \qquad G_2(s) = \frac{1}{m_1}\frac{\omega_{2n}^2}{|\Delta(i\omega)|}$$

$$|\Delta(i\omega)| = \left(-\omega^2 + (\omega_{n1}^2 + \mu\omega_{n2}^2)\right)(-\omega^2 + \omega_{n2}^2) - \mu\omega_{n2}^4$$

$$x(t) = \begin{bmatrix} \chi_1(\omega)p_0\cos(\omega t + \phi_1(\omega)) \\ \chi_2(\omega)p_0\cos(\omega t + \phi_2(\omega)) \end{bmatrix}$$

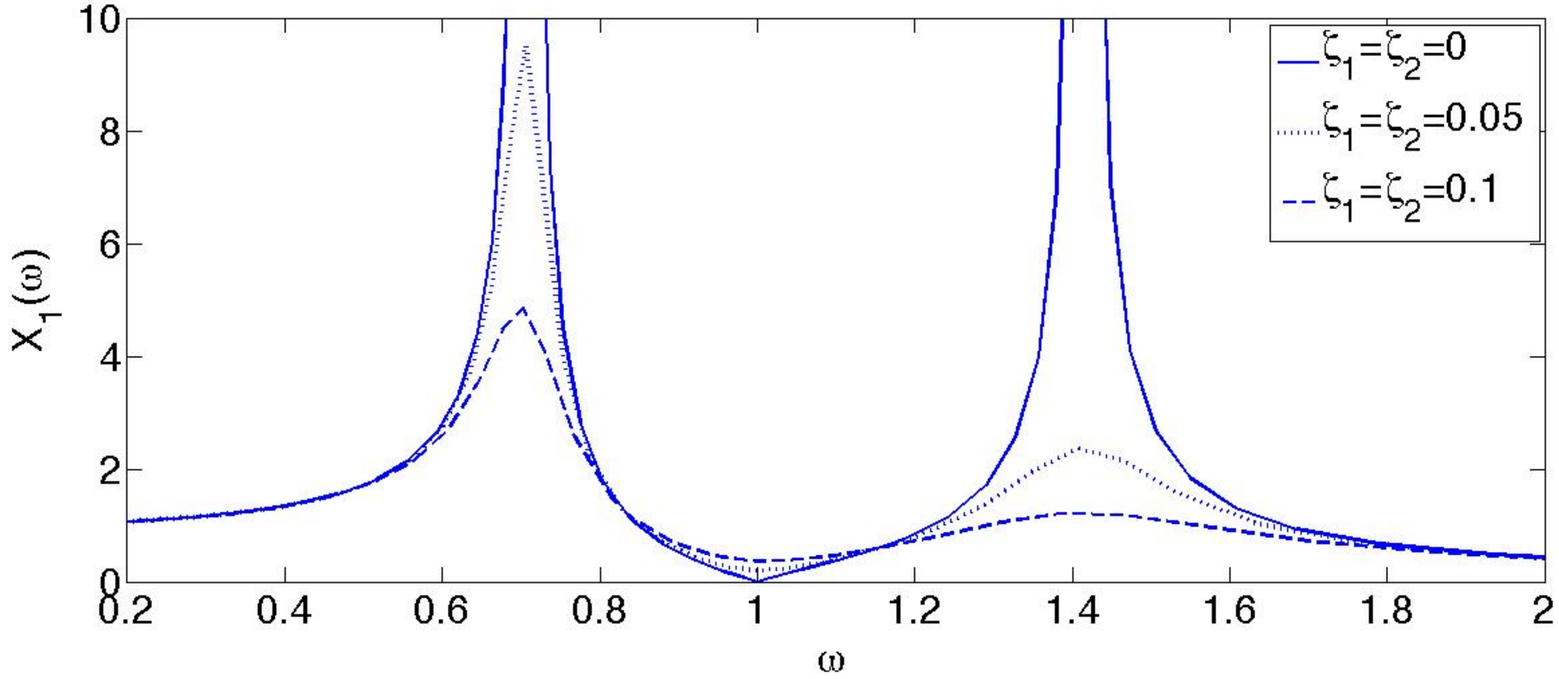


2-DOF Vibrations

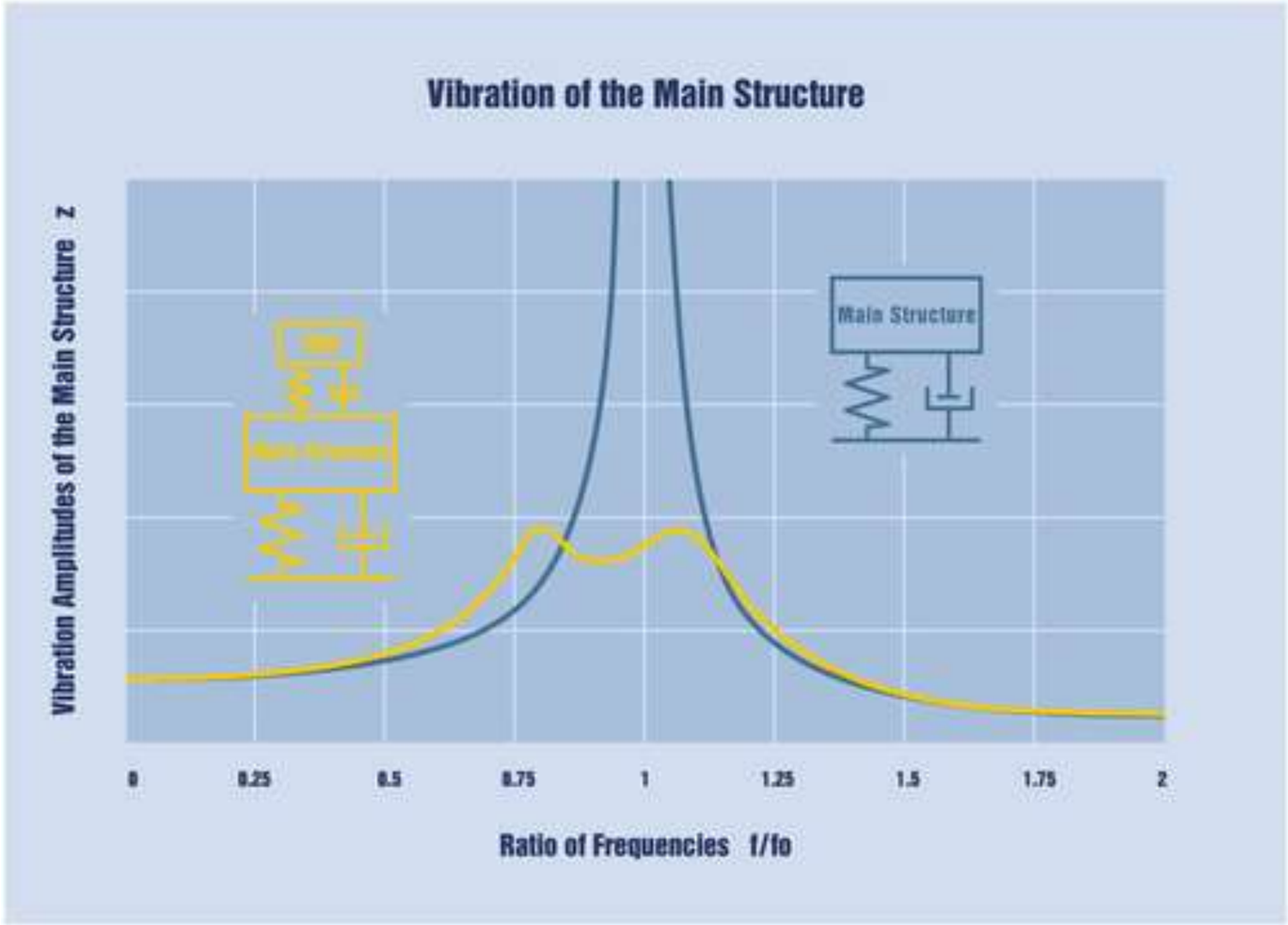
$$\begin{bmatrix} x_1(t) \\ x_2(t)) \end{bmatrix} = \begin{bmatrix} \chi_1(\omega) p_0 \cos(\omega t + \phi_1(\omega)) \\ \chi_2(\omega) p_0 \cos(\omega t + \phi_2(\omega)) \end{bmatrix} \qquad \begin{aligned} \chi_1(\omega) &= |\, G_1(i\omega)\,| \\ \chi_2(\omega) &= |\, G_2(i\omega)\,| \end{aligned} \qquad \begin{aligned} \phi_1(\omega) &= \angle G_1(i\omega) \\ \phi_2(\omega) &= \angle G_2(i\omega) \end{aligned}$$

$$G_1(s) = \frac{m_2 s^2 + c_2 s + k_2}{|\,\Delta(s)\,|} \qquad G_2(s) = \frac{c_2 s + k_2}{|\,\Delta(s)\,|}$$

$$|\,\Delta(s)\,| = \det(\Delta(s)) = \left(m_1 s^2 + (c_1 + c_2)s + (k_1 + k_2)\right)(m_2 s^2 + c_2 s + k_2) - (c_2 s + k_2)^2$$



Tuned Mass Damper

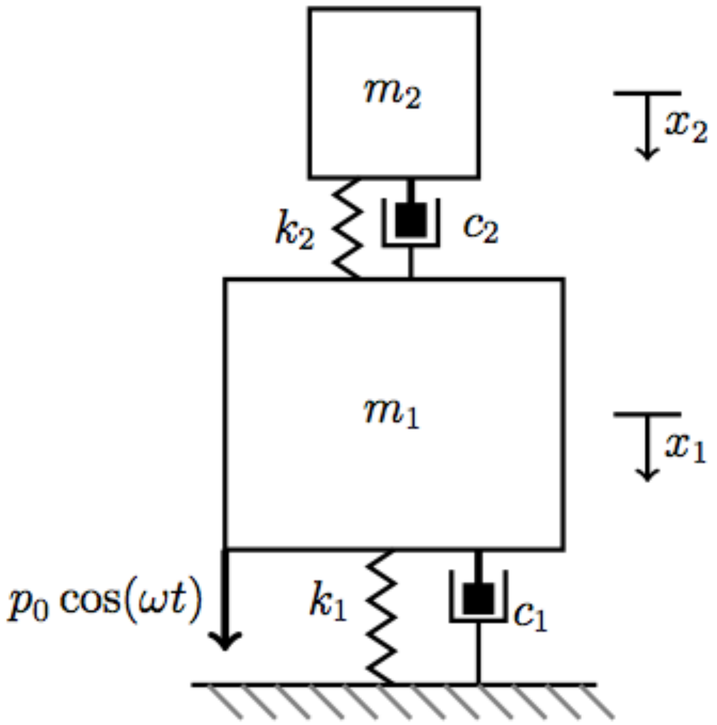


Refer to class note for a detailed discussion

Modal Analysis of a 2DOF System

D. H. S. Maithripala, PhD
Dept. of Mechanical Engineering
University of Peradeniya

Modal Analysis 2-DOF Vibrations



$$M\ddot{x} + C\dot{x} + Kx = f(t)$$

We are typically interested in resonance conditions

Thus let us consider the case $C \approx 0$

$$\ddot{x} + \Omega\, x = M^{-1} f(t)$$

$$\Omega \triangleq M^{-1}K = \begin{bmatrix} (1+\mu\alpha^2)\omega_{n1}^2 & -\mu\alpha^2\omega_{n1}^2 \\ -\alpha^2\omega_{n1}^2 & \alpha^2\omega_{n1}^2 \end{bmatrix}$$

$$\omega_{n1}^2 = \frac{k_1}{m_1} \quad \omega_{n2}^2 = \frac{k_2}{m_2} \quad \mu = \frac{m_2}{m_1} \quad \alpha = \frac{\omega_{n2}}{\omega_{n1}}$$

Resonance Frequencies

Recall that the resonance frequencies ω_1 & ω_2 are given by the roots of

$$|\Delta(i\omega)| = \det\left(-\omega^2 I_{2\times 2} + \Omega\right) \quad \text{where} \quad \Omega \triangleq M^{-1}K = \begin{bmatrix} (1+\mu\alpha^2)\omega_{n1}^2 & -\mu\alpha^2\omega_{n1}^2 \\ -\alpha^2\omega_{n1}^2 & \alpha^2\omega_{n1}^2 \end{bmatrix}$$



$$|\Delta(i\omega)| = \det\left(-\omega^2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} (1+\mu\alpha^2)\omega_{n1}^2 & -\mu\alpha^2\omega_{n1}^2 \\ -\alpha^2\omega_{n1}^2 & \alpha^2\omega_{n1}^2 \end{bmatrix}\right)$$



$$|\Delta(i\omega)| = \det\left(\begin{bmatrix} \left(-\omega^2 + (1+\mu\alpha^2)\right)\omega_{n1}^2 & -\mu\alpha^2\omega_{n1}^2 \\ -\alpha^2\omega_{n1}^2 & (-\omega^2 + \alpha^2\omega_{n1}^2) \end{bmatrix}\right)$$

Hence the square of the resonance frequencies ω_1 & ω_2 are eigenvalues of the matrix Ω

What Do the Eigenvectors Tell Us ?

The corresponding eigen vectors $\bar{x}_1$ & $\bar{x}_2$ are given by the nontrivial solutions of

$$(-\omega_j^2 I + \Omega)\,\bar{x}_j = 0 \quad \text{for} \quad j = 1,2 \;\rightarrow\; \Omega\,\bar{x}_j = \omega^2 \bar{x}_j$$

$$\text{if} \quad T \triangleq [\bar{x}_1 \quad \bar{x}_2] \quad \text{then} \quad \Omega T = TD \quad \text{where} \quad D = \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} = T^{-1}\Omega T$$

$$\text{Let} \quad x = Tz \quad \text{where} \quad z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix}$$

$$\ddot{x} + \Omega\, x = M^{-1} f(t) \quad \rightarrow \quad \begin{bmatrix} \ddot{z}_1 \\ \ddot{z}_2 \end{bmatrix} + \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = T^{-1} \begin{bmatrix} \frac{1}{m_1} f_1(t) \\ 0 \end{bmatrix}$$



What Do the Eigenvectors Tell Us ?

$$\ddot{x} + \Omega\, x = M^{-1} f(t) \quad \rightarrow \quad \begin{bmatrix} \ddot{z}_1 \\ \ddot{z}_2 \end{bmatrix} + \begin{bmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = T^{-1} \begin{bmatrix} \frac{1}{m_1} f_1(t) \\ 0 \end{bmatrix}$$

$$T^{-1} = \begin{bmatrix} T^{11} & T^{12} \\ T^{21} & T^{22} \end{bmatrix}$$

$$\gamma_1 \triangleq \frac{T^{11}}{m_1}, \quad \gamma_2 \triangleq \frac{T^{21}}{m_1}$$

$$\ddot{z}_1 + \omega_1^2 z_1 = \gamma_1 f_1(t)$$
$$\ddot{z}_2 + \omega_2^2 z_2 = \gamma_2 f_1(t)$$

$$x = Tz \quad \rightarrow \; x(t) = \bar{x}_1\, z_1(t) + \bar{x}_2\, z_2(t)$$

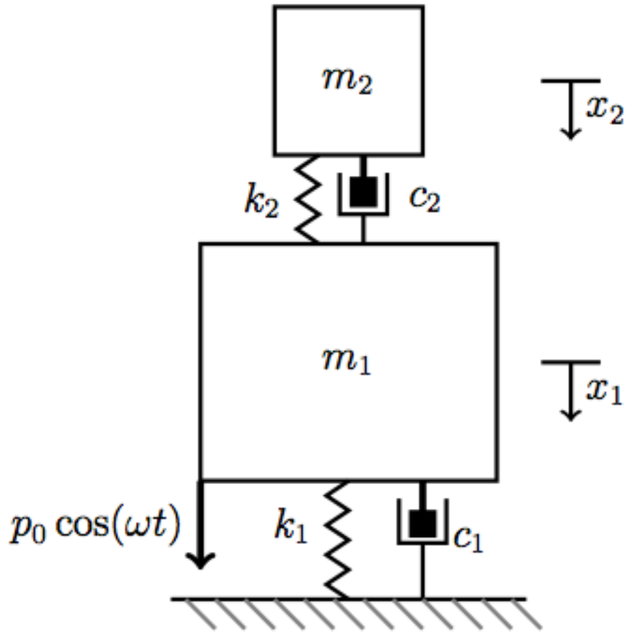
- The response of the system is a linear combination of the response of these two uncoupled systems that we refer to as modes of vibration

- Even though the external force was only acting on the first oscillator in the physical system, it is felt by both uncoupled oscillators in the new coordinates

What Do the Eigenvectors Tell Us ?

$$\ddot{x} + \Omega\, x = M^{-1} f(t) \quad \rightarrow \quad \begin{aligned} \ddot{z}_1 + \omega_1^2 z_1 &= \gamma_1 f_1(t) \\ \ddot{z}_2 + \omega_2^2 z_2 &= \gamma_2 f_1(t) \end{aligned}$$

$$x(t) = \bar{x}_1\, z_1(t) + \bar{x}_2\, z_2(t) \quad \text{where} \quad \bar{x}_j = \begin{bmatrix} \bar{x}_{1j} \\ \bar{x}_{1j} \end{bmatrix} \quad \text{for} \quad j = 1,2$$



$$r_1 \triangleq \frac{\bar{x}_{11}}{\bar{x}_{21}} = \frac{\alpha^2 - 1 - \mu\alpha^2 + \sqrt{(1 + \alpha^2 + \mu\alpha^2)^2 - 4\alpha^2}}{2\alpha^2} > 0$$

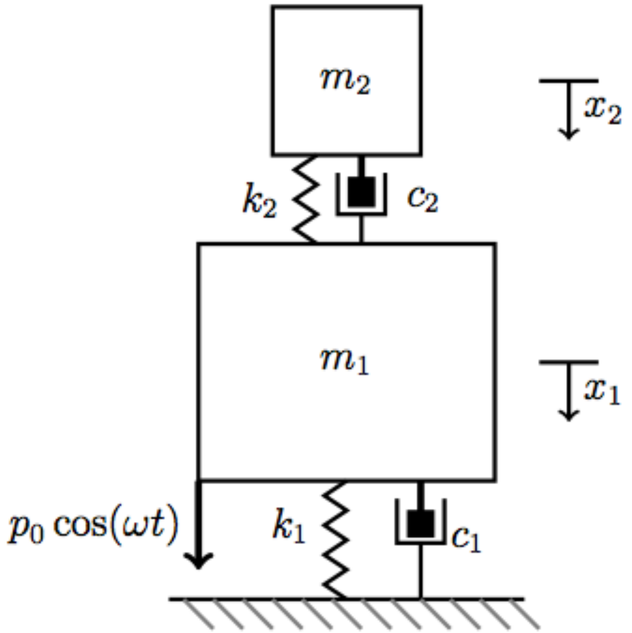
$$r_2 \triangleq \frac{\bar{x}_{12}}{\bar{x}_{22}} = \frac{\alpha^2 - 1 - \mu\alpha^2 - \sqrt{(1 + \alpha^2 + \mu\alpha^2)^2 - 4\alpha^2}}{2\alpha^2} < 0$$

$$\boxed{x(t) = \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} z_1(t) + \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} z_2(t)}$$

Forced Vibration and Resonance

$$\ddot{x} + \Omega\, x = M^{-1} f(t) \quad \rightarrow \quad \begin{aligned} \ddot{z}_1 + \omega_1^2 z_1 &= \gamma_1 f_1(t) \\ \ddot{z}_2 + \omega_2^2 z_2 &= \gamma_2 f_1(t) \end{aligned}$$

Let $\chi_j(\omega) = |G_j(\omega)|$ $\phi_j(\omega) = \angle G_j(\omega)$ where $G_j(\omega) = \dfrac{\gamma_j}{-\omega^2 + \omega_j^2}$.



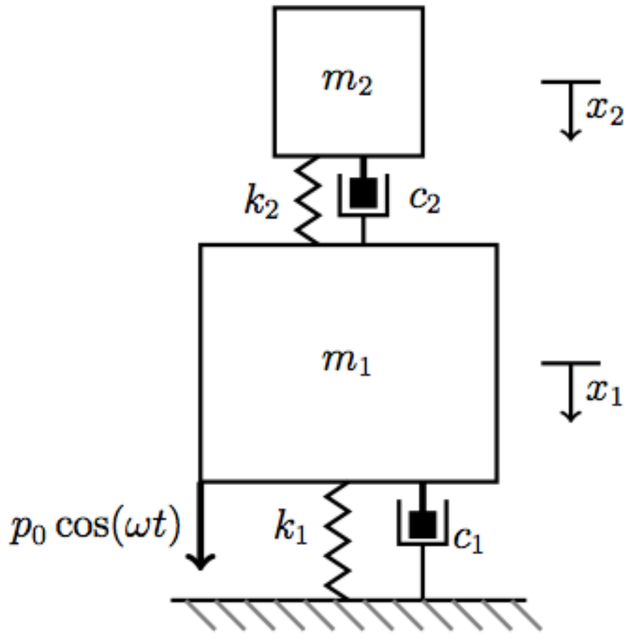
$$z_1(t) = f_0 \chi_1(\omega) \cos(\omega t + \phi_1(\omega))$$
$$z_2(t) = f_0 \chi_2(\omega) \cos(\omega t + \phi_2(\omega))$$

$$x(t) = \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} z_1(t) + \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} z_2(t)$$

Forced Vibration and Resonance

$$\ddot{x} + \Omega\, x = M^{-1} f(t) \quad \rightarrow \quad \begin{aligned} \ddot{z}_1 + \omega_1^2 z_1 &= \gamma_1 f_1(t) \\ \ddot{z}_2 + \omega_2^2 z_2 &= \gamma_2 f_1(t) \end{aligned} \qquad \begin{aligned} z_1(t) &= f_0\, \chi_1(\omega) \cos(\omega t + \phi_1(\omega)) \\ z_2(t) &= f_0\, \chi_2(\omega) \cos(\omega t + \phi_2(\omega)) \end{aligned}$$

Let $\chi_j(\omega) = |\, G_j(\omega)\,| \quad \phi_j(\omega) = \angle G_j(\omega)$ where $G_j(\omega) = \dfrac{\gamma_j}{-\omega^2 + \omega_j^2}$.



$$\boxed{x(t) = \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} z_1(t) + \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} z_2(t)}$$

$$x(t) = f_0\, \chi_1(\omega) \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \cos(\omega t + \phi_1(\omega)) + f_0\, \chi_2(\omega) \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \cos(\omega t + \phi_2(\omega))$$

$$\omega \approx \omega_1 \quad \rightarrow \quad x(t) \approx f_0\, \chi_1(\omega) \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \cos(\omega t + \phi_1(\omega))$$

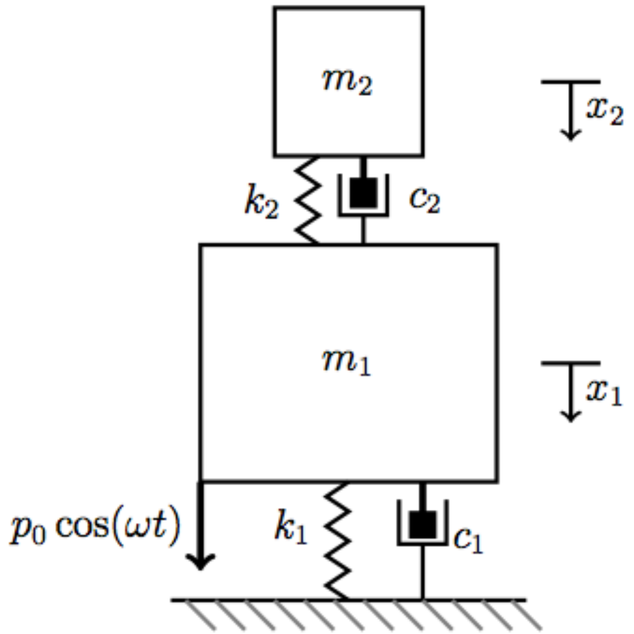
$$\omega \approx \omega_2 \quad \rightarrow \quad x(t) \approx f_0\, \chi_2(\omega) \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \cos(\omega t + \phi_2(\omega))$$

Free Vibrations

$$\ddot{x} + \Omega\, x = 0 \quad \rightarrow \quad \begin{aligned} \ddot{z}_1 + \omega_1^2 z_1 &= 0 \\ \ddot{z}_2 + \omega_2^2 z_2 &= 0 \end{aligned}$$

$$\boxed{x(t) = \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} z_1(t) + \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} z_2(t)}$$

$$z_1(t) = A_1 \cos(\omega_1 t + \phi_1)$$
$$z_2(t) = A_2 \cos(\omega_2 t + \phi_2)$$



$$x(t) = A_1 \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \cos(\omega_1 t + \phi_1) + A_2 \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \cos(\omega_2 t + \phi_2)$$

$$\dot{x}(0) = 0, \; x(0) \approx \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \quad \rightarrow \quad x(t) \approx \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \cos \omega_1 t$$

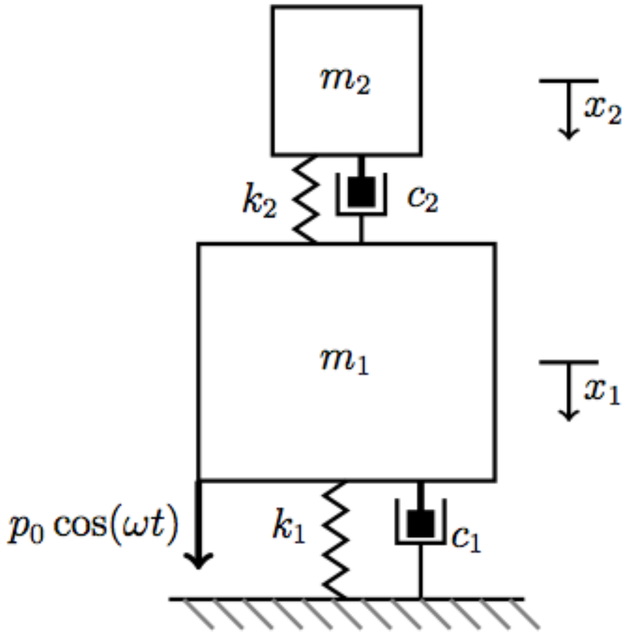
$$\dot{x}(0) = 0, \; x(0) \approx \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \quad \rightarrow \quad x(t) \approx \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \cos \omega_2 t$$

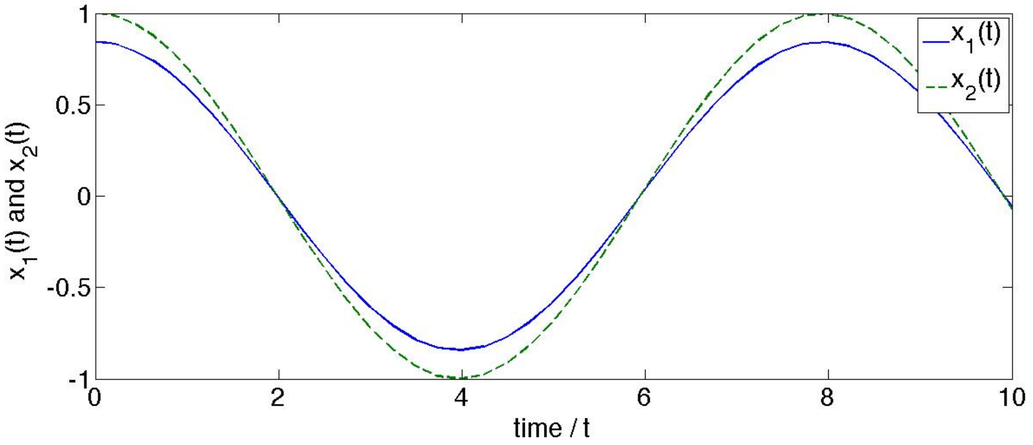
Free Vibrations

$$\ddot{x} + \Omega x = 0 \quad \rightarrow \quad \begin{aligned} \ddot{z}_1 + \omega_1^2 z_1 &= 0 \\ \ddot{z}_2 + \omega_2^2 z_2 &= 0 \end{aligned}$$

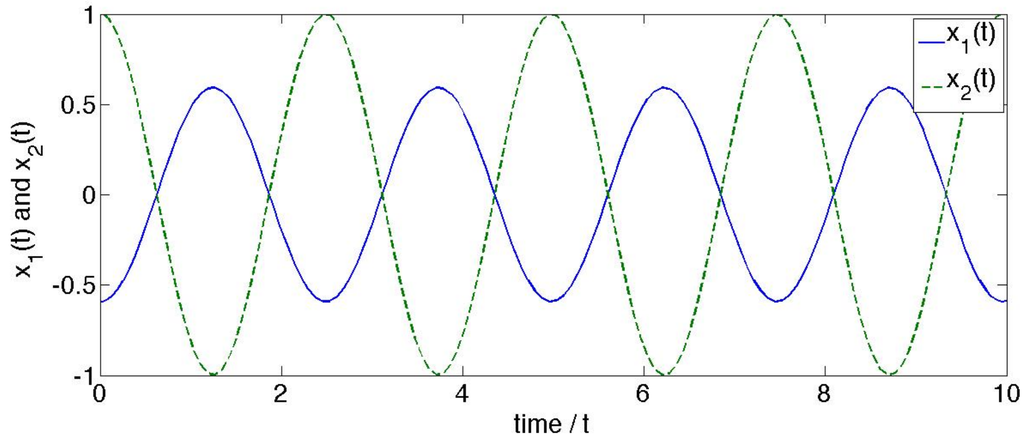
$$x(t) = \bar{x}_{21} \begin{bmatrix} r_1 \\ 1 \end{bmatrix} z_1(t) + \bar{x}_{22} \begin{bmatrix} r_2 \\ 1 \end{bmatrix} z_2(t)$$

$$z_1(t) = A_1 \cos(\omega_1 t + \phi_1)$$
$$z_2(t) = A_2 \cos(\omega_2 t + \phi_2)$$





$$\dot{x}(0) = 0, \; x(0) \approx \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \quad \rightarrow \quad x(t) \approx \begin{bmatrix} r_1 \\ 1 \end{bmatrix} \cos \omega_1 t$$

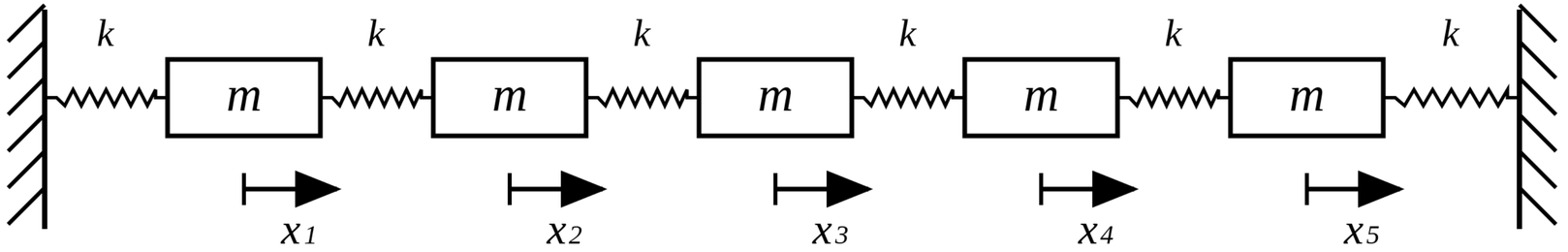


$$\dot{x}(0) = 0, \; x(0) \approx \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \quad \rightarrow \quad x(t) \approx \begin{bmatrix} r_2 \\ 1 \end{bmatrix} \cos \omega_2 t$$

Modal Analysis in General

D. H. S. Maithripala, PhD
Dept. of Mechanical Engineering
University of Peradeniya

Many Degrees of Freedom System



$$M\ddot{x} + Kx = f(t), \qquad x(t) \in \mathbb{R}^n \quad \Rightarrow \quad \ddot{x} + \Omega x = M^{-1}f(t), \qquad \Omega = M^{-1}K$$

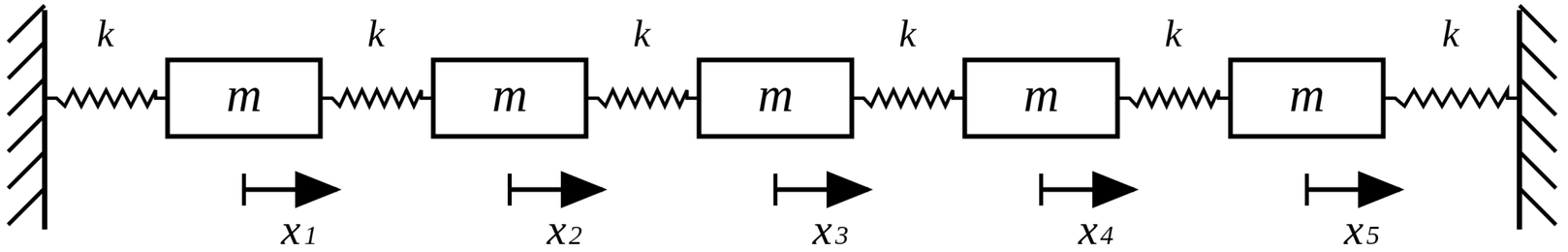
Define the Inner Product: $\langle\langle x, y\rangle\rangle_M = y^T M x$

$$\langle\langle \Omega x, y\rangle\rangle_M = y^T K x = \langle\langle x, \Omega y\rangle\rangle_M \quad \Rightarrow \quad \Omega\bar{x}_j = \omega_j^2 \bar{x}_j, \qquad \langle\langle \bar{x}_i, \bar{x}_j\rangle\rangle_M = \delta_{ij}$$

Typically $\lambda(\Omega) = \{\omega_1^2, \cdots, \omega_n^2\}$ are distinct $\Rightarrow$ $\{\bar{x}_1, \cdots, \bar{x}_n\}$ orthogonally span $\mathbb{R}^n$

$$x(t) \in \mathbb{R}^n \quad \Rightarrow \quad x(t) = \sum_{k=1}^{n} z_k(t)\bar{x}_k, \qquad \langle\langle \bar{x}_k, x(t)\rangle\rangle_M = \sum_{j=1}^{n} z_j(t)\langle\langle \bar{x}_k, \bar{x}_j\rangle\rangle_M = z_k(t)$$

Many Degrees of Freedom System



$$\ddot{x} + \Omega x = M^{-1} f(t), \qquad \Omega = M^{-1} K, \qquad \lambda(\Omega) = \{\omega_1^2, \cdots \omega_n^2\}$$

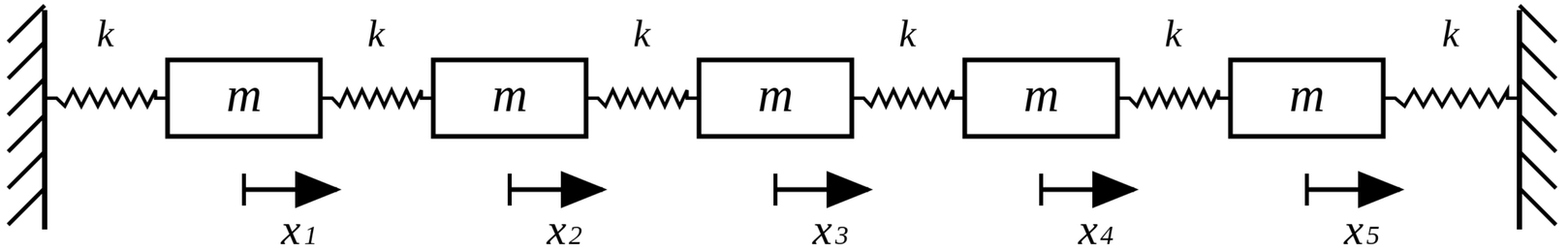
$$x(t) = \sum_{i=1}^{n} z_i(t) \bar{x}_i, \qquad \ddot{x}(t) = \sum_{i=1}^{n} \ddot{z}_i(t) \bar{x}_i$$

$$\langle\langle \bar{x}_k, \Omega x(t) \rangle\rangle_M = \sum_{j=1}^{n} z_j(t) \langle\langle \bar{x}_k, \Omega \bar{x}_j \rangle\rangle_M = \omega_k^2 z_k(t), \qquad \langle\langle \bar{x}_k, \ddot{x}(t) \rangle\rangle_M = \sum_{j=1}^{n} \dot{z}_j(t) \langle\langle \bar{x}_k, \bar{x}_j \rangle\rangle_M = \ddot{z}_k(t)$$

$$\sum_{k=1}^{n} \left(\ddot{z}_k + \omega_k^2 z_k \right) \bar{x}_k = \sum_{k=1}^{n} \gamma_k(t) \bar{x}_k, \qquad \gamma_k(t) = \langle\langle \bar{x}_k, M^{-1} f(t) \rangle\rangle = \bar{x}_k^T f(t)$$

$$\ddot{z}_k + \omega_k^2 z_k = \gamma_k(t), \qquad k = 1, \ldots, n$$

Many Degrees of Freedom System



$$\ddot{x} + \Omega x = M^{-1} f(t), \qquad \Omega = M^{-1} K, \qquad \lambda(\Omega) = \{\omega_1^2, \cdots \omega_n^2\}$$

$$x(t) = \sum_{i=1}^{n} z_i(t) \bar{x}_i, \qquad \gamma_k(t) = \langle\langle \bar{x}_k, M^{-1} f(t) \rangle\rangle = \bar{x}_k^T f(t)$$

$$\ddot{z}_k + \omega_k^2 z_k = \gamma_k(t), \qquad k = 1, \ldots, n \qquad z_k(0) = \langle\langle \bar{x}_k, x(0) \rangle\rangle_M, \qquad \dot{z}_k(0) = \langle\langle \bar{x}_k, \dot{x}(0) \rangle\rangle_M$$

$$f(t) = f_0 \cos(\omega t) \quad \Rightarrow \quad b_k = \bar{x}_k^T f_0$$

$$\omega \neq \omega_k \quad \Rightarrow \quad z_k(t) = z_k(0)\cos(\omega_k t) + \frac{\dot{z}_k(0)}{\omega_k} \sin(\omega_k t) + \frac{b_k}{\omega_k^2 - \omega^2} \left(\cos(\omega t) - \cos(\omega_k t)\right)$$

$$\omega = \omega_k \quad \Rightarrow \quad z_k(t) = z_k(0)\cos(\omega_k t) + \frac{\dot{z}_k(0)}{\omega_k} \sin(\omega_k t) + \frac{b_k}{2\omega_k} t \sin(\omega_k t)$$

Many Degrees of Freedom System

Effect of Damping

$$\ddot{x} + M^{-1}C\dot{x} + \Omega x = M^{-1}f(t), \qquad \Omega = M^{-1}K, \qquad f(t) = f_0 \cos(\omega t)$$

$$x(t) = \sum_{k=1}^{n} z_k(t)\bar{x}_k, \qquad \ddot{z}_k + \sum_{k=1}^{n} 2c_{jk}\dot{z}_k + \omega_k^2 z_k = \bar{x}_k^T f_0, \qquad c_{jk} = \langle\langle \bar{x}_j, M^{-1}C\bar{x}_k\rangle\rangle_M = \bar{x}_j^T C\bar{x}_k$$

$$C = bK \quad \Rightarrow \quad c_{jk} = \langle\langle \bar{x}_j, M^{-1}C\bar{x}_k\rangle\rangle_M = 2\zeta_k\omega_k\delta_{jk}$$

$$\omega_{dk} = \omega_k\sqrt{1-\zeta_k^2}$$

$$0 \le \zeta < 1 \quad \Rightarrow \quad z_k(t) = e^{-\zeta_k\omega_k t}\left[z_k(0)\cos(\omega_{dj}t) + \frac{\dot{z}_k(0) + \zeta_j\omega_k z_k(0)}{\omega_{dk}}\sin(\omega_{dk}t)\right] + \int_0^t \frac{e^{-\zeta_k\omega_j(t-\tau)}}{\omega_{dk}}\sin\big(\omega_{dk}(t-\tau)\big)\gamma_k(\tau)\,d\tau$$

Continuous System

The Basic Idea of Fourier Series

n-dimensional complex vector space: $\mathbb{C}^n$

Define the Inner Product: $\langle\langle x, y \rangle\rangle = y^* x = \sum_{j=1}^{n} \overline{x}_j y_j$

Orthonormal basis for $\mathbb{C}^n \quad \Rightarrow \quad \{\psi_1, \ldots, \psi_n\} \quad \Rightarrow \quad \langle\langle \psi_k, \psi_l \rangle\rangle = \delta_{kl}$

$$\text{Every} \quad x \in \mathbb{C}^n \quad \Rightarrow \quad x = \sum_{k=1}^{n} \alpha_k \psi_k \quad \text{where} \quad \alpha_k = \langle\langle \psi_k \, , x \rangle\rangle$$

Continuous System

The Basic Idea of Fourier Series

Infinite dimensional vector space over: $\mathbb{C}^n \quad \Rightarrow \quad \mathcal{F} = L^2(0,L;\mathbb{C})$

$$f \in \mathcal{F} \quad \Rightarrow \quad \int_0^L |f(x)|^2 \, dx < \infty$$

Define the Inner Product: $\langle\langle f, g \rangle\rangle = \frac{1}{L} \int_0^L f(x) \overline{g(x)} \, dx, \quad \forall f, g \in \mathcal{F}$

$$\psi_k(x) = e^{i2\pi kx/L}, \qquad k \in \mathbb{Z}$$

Orthonormal basis for $\mathcal{F} \quad \Rightarrow \quad \{\psi_1, \psi_2 \ldots\} \quad \Rightarrow \quad \langle\langle \psi_k, \psi_l \rangle\rangle = \delta_{kl}$

Every $f \in \mathcal{F} \quad \Rightarrow \quad f = \sum_{-\infty}^{\infty} \alpha_k \psi_k = \underbrace{\sum_{-\infty}^{\infty} \alpha_k e^{i2\pi kx/L}}_{Fourier\ Series}$ where $\alpha_k = \langle\langle \psi_k \, f \rangle\rangle = \underbrace{\frac{1}{L} \int_0^L f(x) e^{-i2\pi kx/L} \, dx}_{Fourier\ Coefficient}$

Continuous System

$$\frac{\partial^2 y}{\partial t^2} + C\frac{\partial y}{\partial t} + Hy = q(\mathbf{x}, t), \qquad y(\,\cdot\,, \mathbf{x}) \in \mathscr{F} = L^2(\mathscr{D}, w\, d\mathbf{x})$$

Define the Inner Product: $\langle\langle u, v \rangle\rangle = \int_{\mathscr{D}} w(\mathbf{x}) \overline{u(\mathbf{x})} v(\mathbf{x})\, d\mathbf{x}$

$$\langle\langle Hu, v \rangle\rangle = \langle\langle u, Hv \rangle\rangle \quad \Rightarrow \quad H\psi_k = \omega_k^2 \psi_k, \qquad \psi_k \in \mathscr{F}, \quad \langle\langle \psi_k, \psi_l \rangle\rangle = \delta_{kl}, \quad \omega_k \in \mathbb{R}$$

$$y(\mathbf{x}, t) = \sum_{k=1}^{\infty} \alpha_k(t) \psi_k(\mathbf{x}), \qquad \alpha_k(t) = \langle\langle \psi_k\,, y(\,\cdot\,, t) \rangle\rangle$$

$$\ddot{\alpha}_k + \sum_{j=1}^{\infty} c_{kj} \dot{\alpha}_j + \omega_k^2 \alpha_k = f_k(t), \qquad c_{kj} = \langle\langle \psi_k, C\psi_j \rangle\rangle, \qquad f_k(t) = \langle\langle \psi_k\,, q(\,\cdot\,, t) \rangle\rangle$$

$$C = cI \quad \Rightarrow \quad \ddot{\alpha}_k + 2\zeta_k \omega_k \dot{\alpha}_k + \omega_k^2 \alpha_k = f_k(t), \qquad 2\zeta_k \omega_k = c$$

Thank You